BRGF	**TMIRI Fall 2025** **Delivery date: 07/11/2025** **Student Name: Volpi Davide**

1. Gemini

- Choose a topic and using Gemini, ask for 5 references on that topic: (A typical formulation of a question would be: "*Please, provide references on Optical Constellation Slicing"*)
- Check the references obtained with Gemini using Google Scholar, the webpages of the publisher, or other sources

List here the 5 references proposed by Gemini	Are the references indexed in Google Scholar?	Are the references in the web of the publisher or somewhere else?
1. AI MARRVEL: A new AI tool to diagnose genetic disorders	No	Texas Children's Hospital
2.Artificial intelligence-based approaches for the detection and prioritization of genomic mutations in congenital surgical diseases	Yes	PubMed Central
3.Building Machine Learning Challenges for Anomaly Detection in Science	Yes	ResearchGate
4.Genomic Anomaly Detection with Functional Data Analysis	Yes	MDPI
5.AI for Embryo Grading and Pregnancy Prediction	No	Bioengineering.org

- Ask Gemini about the sources used to extract the previous question and the criteria of selection. Copy and paste the answers
  - The sources used to extract the previous list of references were the results of two distinct Google searches:
    -AI for genetical anomaly detection references

-machine learning genomics anomaly detection papers
Criteria for Selection:
-The primary criteria for selecting the references that were presented to you were based on the relevance of the search snippets to the user's request for "AI for Genetical Anomaly detection":
-Keyword Match: Snippets explicitly mentioning "AI," "Machine Learning (ML)," "Deep Learning (DL)," and "genomic," "genetic," "mutations," "disorders," or "anomalies" were prioritized (e.g., references 1.2, 1.3, 1.7, 2.5, 2.6, 2.9).
-Application Area: References discussing the application of AI in related fields like fetal anomaly detection via genomics (e.g., Down Syndrome diagnosis from genetic data) or protein anomaly detection based on genomic sequences were included as they represent a closely related domain (e.g., references 1.4, 1.6, 1.10).
-General Anomaly Detection in Genomics: Papers that discussed using ML/AI for general anomaly detection within the field of genomics (even if not strictly clinical genetics) were also considered relevant (e.g., references 2.3, 2.5).
-The results from the second search, "machine learning genomics anomaly detection papers," primarily provided more technical papers on the general methodology of anomaly detection in scientific data, including genomics, which helped broaden the selection to include the more fundamental algorithmic approaches (e.g., references 2.1, 2.3).
The final list was a curated selection of papers where the connection between AI/ML and the detection of genetic or genomic anomalies was clearly established in the snippet.

- Evaluate the experience
  - The experience was very positive. Gemini provided a comprehensive list of references that included the title, author, direct link, and a brief description for each source. The response was concise, accurate, and appropriate for the initial stage of research. Considering that the original question was broad in scope, the reasoning behind the selection of sources appears appropriate and methodologically sound for a preliminary exploration of the topic.

2. INSPEC

Bibliotècnica (http://bibliotecnica.upc.edu/) > Bases de dades > Inspec (*Engineering Village*)
Before install the button eBIB in your navigator.

- Choose a topic of your interest.
- Write the most appropriate search strategy to find information on your topic. Then, submit your search in INSPEC database.
- Use the thesaurus if you need it to find keywords. Analyze all the results retrieved, whether they are appropriate or not for your search query. Modify the search if you need to find relevant information, without noise or silence.
- If you don't find information about your topic in INSPEC database, just change your topic and search again in INSPEC.

Topic of your research: AI for genetic anomaly detection
Please copy here your search string: ("artificial intelligence" OR "machine learning" OR "deep learning") AND ("genetic anomaly" OR "genetic disorder" OR "genome analysis" OR "mutation detection" OR genom*)
Description of your search strategy: *- Have you used keywords from the thesaurus? Which ones? Why/Why not?* *- Did you use Boolean operators, truncation symbols, or square brackets in your search strategy?* *- Did you use any options to limit the search results? For example, by language, dates, document type... Explain and justify why you used them.* I checked the INSPEC Thesaurus to identify controleld terms such as artificial intelligence, machine learning and genomics. However, since the term "genetic anomaly detection" is not a standard thesaurus term, I combined the previously mentioned terms with other language keywords such as "mutation detection" to capture a bigger range of articles. I used OR to include synonyms and related terms, and AND to intersect AI related and genetic related concepts, with parentheses to structure the logic of the query. I've used also truncation to include variations like "genome" and "genomic". I used some limitation in the date range (2015-2025) to limit the articles only to recent and relevant research, the language only to english to ensure interpretability, document type as journal and conference, subject area filters to focus results on intersection of AI, bioinformatics and biomedical research.
Results evaluation: *- Are all retrieved results suitable and relevant? Why?* *- How have you refined or modified your search query to improve retrieved results?* Not all retrieved results were fully relevant. While many articles focused on the use of AI/ML for genomic or mutation analysis, some addressed general AI applications in healthcare or bioinformatics without specifically targeting genetic anomaly detecction. I was expecting this because general terms like "genomic analysis" or "genomics" can retrieve studies on different topics, related but not directly about anomaly detection. To improve the precision of the query I've tried to add more specific keywords such as "disease prediction" and "variant classification" to focus on detection task, I've added the term "bioinformatics" to capture more domain-specific research. The new query became: ("artificial intelligence" OR "machine learning" OR "deep learning") AND ("genetic anomaly detection" OR "mutation detection" OR "variant classification" OR "disease prediction") AND bioinformatics

3. Web of Science

- Access to Web of Science

1. **Perform the following basic search at Web of Science – Core Collection- in the topic *Cybersecurity in electrical power networks*. Consider different ways to name each concept. Once you have the results, answer the following questions:**

   a. Please copy here your search string
      i. (("cyber security" OR cybersecurity OR "information security") AND ("electrical power network*" OR "smart grid*" OR "power system*" OR "electric grid"))

   b. How many results are retrieved?
      i. 2836

   c. Who are the three authors with the most published works?
      i. Wang LF, Govindarasu M, Liu CC

   a. What are the five organizations (affiliations) with the most published works?

UNITED STATES DEPARTMENT OF ENERGY DOE, STATE UNIVERSITY SYSTEM OF FLORIDA, STATE GRID CORPORATION OF CHINA, INDIAN INSTITUTE OF TECHNOLOGY SYSTEM IIT SYSTEM, TEXAS A M UNIVERSITY SYSTEM

   d. How many documents are authored by researchers affiliated with Spanish institutions?
      i. 56

   e. Which is the first country in the article publication ranking?
      i. USA with 924

   f. Which is the title of the publication (journal, conference, book, etc.) Which of them published more articles?
      i. IEEE Access with 154

   g. What is the journal article, published since 2020, that has received the most citations so far?
      i. Artificial intelligence in sustainable energy industry: Status Quo, challenges and opportunities (534 citations)

2. **Look for documents by author Jonatas Boas Leite in Web of Science Core Collection. For each question, please explain how you have done your search**

   a. How many of his works have been indexed in Web of Science?
      i. After going to the WoS Core Collection where you research for Researchers, I've started my research as "Name Search" – Last Name "LEITE" and First Name "JONATAS BOAS" and then clicked on search. On the right of the resulting page, under the Profile Summary section, the number of works indexed in WoS can be found near the "Publications indexed in WoS" part, which in this case is 41.

b. Where do you currently work according to WoS?
    i. In the middle of the resulting page, under “Organizations”, the current work can be found, and in this case he is affiliaed with the Universidade Estadual Paulista and Instituto Politecnico do Porto.

c. Which is his most-cited document? How many citations?
    i. In the section Documents, is possible to filter for Citations-highest to lower. The resulting rank shows that the first article for citations is “Detecting and Locating Non-Technical Losses in Modern Distribution Networks” with 110 citations.